

Lesson 11: Traversals Practice

45 minutes

Overview

Students practice traversing lists, filtering and reducing lists, and using the data import tools. Along the way students develop debugging practices with traversals.

Standards

Full Course Alignment

CSP Conceptual Framework

- **AAP-2** - The way statements are sequenced and combined in a program determines the computed result. Programs incorporate iteration and selection constructs to represent repetition and make decisions to handle varied input values.

CSTA K-12 Computer Science Standards (2017)

- **AP** - Algorithms & Programming

Agenda

Warm Up (2 minutes)

Quick Warm Up

Activity (38 minutes)

Practice Time

Traversal Practice

Reduce and Filter Practice

App Practice

Wrap Up (5 minutes)

Synthesizing Discussion

Assessment: Check For Understanding - AP Practice

Objectives

Students will be able to:

- Debug programs that use list traversals
- Write programs that use list traversals, including the filter and reduce patterns, with the support of sample code

Preparation

- Review the levels that students will be completing with an eye for how you will encourage them to use the debugging practices emphasized in today's lesson.
- Check the **"Teacher's Lounge"** forum for verified teachers to find additional strategies or resources shared by fellow teachers
- If you are teaching virtually, consider checking our **Virtual Lesson Modifications**

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the teachers

- **CSP Unit 6 - Lesson 11 - Traversals Practice** - Slides [▼ Make a Copy](#)

Teaching Guide

Warm Up (2 minutes)

Quick Warm Up

Remarks

Today we're going to have a chance to practice programming with a lot of the concepts and patterns we've explored over the last two lessons. I encourage you to go through these with a partner, but pay close attention to what each other is doing. In our next lesson you're going to have to use a lot of these on an independent project, and these activities are good practice for what you'll find there! Alright, let's get to it!

Teaching Tip

Move Quickly to the Activity: There's a lot in the main activity of today's lesson. You may optionally wish to do a quick vocabulary review or address any questions that came up in the last lesson. Otherwise, give students more time to get hands on with some code.

Activity (38 minutes)

Practice Time

Group: It is recommended that students work in pairs for this lesson and a number of the activities feature discussion prompts. Consider using pair programming, having drivers and navigators switch every 3 minutes, not every level.

Remarks

 Today you're mostly going to practice what we've learned about programming with traversals. I'm here to help you when you need. However, I first want to remind you of the following:

- Use your debugging skills. Try to zoom in on precisely where you're getting stuck.
- Talk to your partner! That's what they're there for!
- Hover over blocks to read the documentation about how they work.
- Read the resources in the Help & Tips tab
- Talk to the group next to you. If another group asks for help make sure to take some time to talk it through with them.

 We can debug traversals by using many skills that helped us with lists like using the watch panel and console.log. Another important new debugging skill will be to actually go look at your data in the Data tab. Use console.log to make sure you're successfully getting the data you want.

Traversal Practice

These levels have students perform simple traversals over lists that are created inside their code (not with the data import tools). In each program sample code is provided which students can use to help writing the code of their own.

- **Level 1:** traverse over a list and console.log every element
- **Level 2:** traverse over a list and console.log every element and its position

- **Level 3:** traverse over two parallel lists and print elements from both



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Traversal Practice

1

2

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💡 Teaching Tip

Providing Support: Circulate around the room through the lesson encouraging students to use the strategies introduced at the beginning of the lesson. Students have a number of supports at their fingertips, so a big part of your role is helping build their independence in using those resources.

Reduce and Filter Practice

Students practice the reduce and filter patterns introduced in the previous lesson. In each program sample code is provided which students can use to help writing the code of their own.

- **Level 4:** filter a list of students to keep only those with more than 6 letters in their names.
- **Level 5:** reduce a list to find the maximum price inside of it.
- **Level 6:** filter a list of countries to find those in the "Central America" region. This program makes use of the data import tools.



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Reduce and Filter Practice

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5

6

App Practice

- **Level 7:** Choice level where students use the reduce and filter patterns to finish building an app.
 - Level 7a - Students add code to calculate a user's average grade.
 - Level 7b - Students add code to allow a user to look up Air Quality data from their state.
 - Level 7c - Students update a program so that it shows average gas prices for multiple cities across the USA.



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Putting It Together

💡 Teaching Tip

Choice Level: Consider polling your students to see which level they would like to review as a class.

Wrap Up (5 minutes)

Synthesizing Discussion

 **Discuss:** *What aspects of working with traversals do you feel like clicked today? What do you still feel like you have trouble with?*

Discussion Goal: Use this opportunity to address any lingering questions or misconceptions in the room. You can also use this as a source of discussion topics to kick off the following lesson. As you lead the discussion, call out the many resources students have access to help when they're getting stuck.

Remarks

Working with traversals can be tricky. We will get more practice tomorrow by making an app that uses traversals and the data tools.

Assessment: Check For Understanding - AP Practice

Check For Understanding Question(s) and solutions can be found in each lesson on Code Studio. These questions can be used for an exit ticket.

Question: What is stored in `studentScores` after running the program code?

```
studentScores ← [77, 32, 45, 92, 86]

FOR EACH item IN studentScores
{
  IF (item > 60)
  {
    item ← item + 5
  }
  ELSE
  {
    item ← 0
  }
}
```

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Check For Understanding: AP Practice



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